



# Método de la Secante

## Introducción

Cuando se tiene una función "complicada", uno de los *inconvenientes* del método de Newton estriba en que se requiere conocer la primera derivada de la función para ser evaluada en diferentes puntos, una de las posibilidades para lograr esto es emplear el método de la secante.

## Fundamentos

$$x_{k+1} = x_k - \frac{f(x_k)}{f'(x_k)} \quad \dots\dots\dots (1)$$

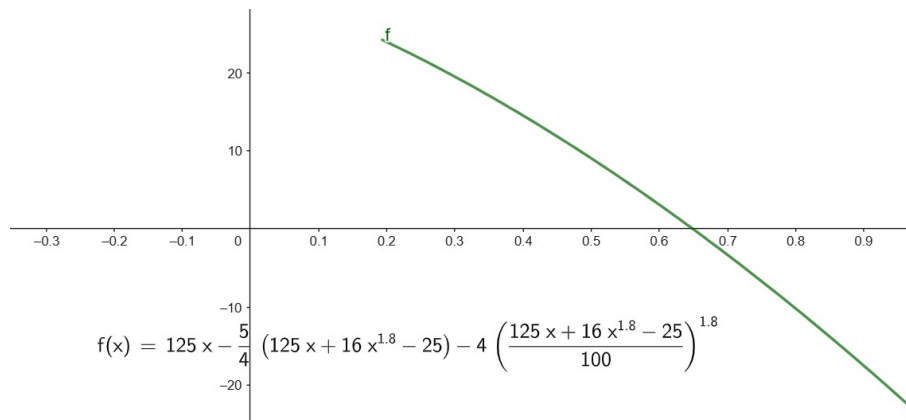
$$f'(x_k) = \frac{f(x_k) - f(x_{k-1})}{x_k - x_{k-1}} \quad \dots\dots\dots (2)$$

(2) en (1)

$$x_{k+1} = x_k - \frac{f(x_k)}{\frac{f(x_k) - f(x_{k-1})}{x_k - x_{k-1}}} \quad \dots\dots\dots x_{k+1} = x_k - \frac{f(x_k) \cdot (x_k - x_{k-1})}{f(x_k) - f(x_{k-1})}$$



$$f(x) := 125x - \frac{5}{4} \cdot (125x + 16x^{1.8} - 25) - 4 \cdot \left( \frac{125x + 16x^{1.8} - 25}{100} \right)^{1.8}$$



$$S(f, 0.6, 0.8) = \begin{bmatrix} 1 & 0.646737187093436 & 0.163743947632698 \\ 2 & 0.64916700719508 & 8.35190766335847 \cdot 10^{-3} \\ 3 & 0.649297603544363 & -7.69014555346281 \cdot 10^{-6} \\ 4 & 0.649297483406414 & 3.60419027956027 \cdot 10^{-10} \\ 5 & 0.649297483412044 & 9.54791801177635 \cdot 10^{-15} \end{bmatrix}$$

$$x = 0.649297483412044$$



2. Utilizando el método de la secante calcular el volumen específico del n-butano utilizando la ecuación de estado propuesta por Patel Teja, con los datos proporcionados

$$P = \frac{R \cdot T}{V - b} - \frac{a}{V \cdot (V + b) + c \cdot (V - b)}$$

Donde:

$$a = \varphi_A \cdot \left( \frac{R^2 \cdot T_C}{P_C} \right) \cdot (1 + F \cdot \sqrt{Tr})^2 \quad b = \varphi_B \cdot \left( \frac{R \cdot T_C}{P_C} \right) \quad c = \varphi_C \cdot \left( \frac{R \cdot T_C}{P_C} \right)$$

$$F = 0.452413 + 1.30982 w - 0.295937 w^2 \quad \lambda_C = 0.329 - 0.0768 w + 0.0212 w^2$$

$$\varphi_A = 3 \cdot \lambda_C^2 + 3 \cdot (1 - 2 \cdot \lambda_C) \cdot \varphi_B + \varphi_B^2 + 1 - 3 \cdot \lambda_C$$

sabiendo  $\varphi_B$  que es el valor positivo mas pequeño

$$\varphi_B^3 + (2 - 3 \cdot \lambda_C) \cdot \varphi_B^2 + 3 \cdot \lambda_C^2 \cdot \varphi_B - \lambda_C^3 = 0 \quad \varphi_C = 1 - 3 \cdot \lambda_C$$

Datos adicionales:

$$w = 0.1850 \quad T_C = 440.5 \cdot K \quad T = 480 \cdot K \quad Tr = \frac{T}{T_C}$$

$$P_C = 37.473 \cdot \text{Atm} \quad R = 0.082 \cdot \frac{\text{Atm} \cdot L}{\text{mol} \cdot K} \quad P := 1$$

**Solución**

$$w := 0.185 \quad R := 0.082 \quad T_C := 440.5 \quad P_C := 37.473$$

$$\lambda_C := 0.329 - 0.0768 w + 0.0212 w^2 \quad \lambda_C = 0.31551757$$

$$\varphi_C := 1 - 3 \cdot \lambda_C \quad \varphi_C = 0.05344729$$

$$c := \varphi_C \cdot \left( \frac{R \cdot T_C}{P_C} \right) \quad c = 0.0515189486321$$

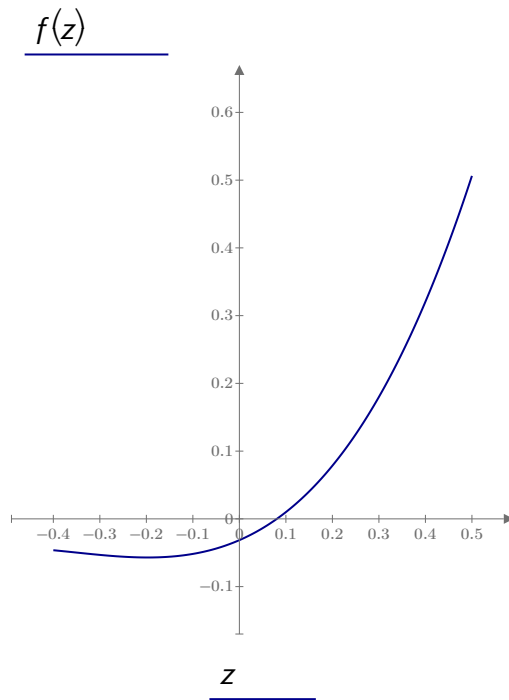


$$\varphi_B^3 + (2 - 3 \cdot \lambda_C) \cdot \varphi_B^2 + 3 \cdot \lambda_C^2 \cdot \varphi_B - \lambda_C^3 = 0$$

$$T := 480$$

$$P_C := 37.473$$

$$f(\varphi_B) := \varphi_B^3 + (2 - 3 \cdot \lambda_C) \cdot \varphi_B^2 + 3 \cdot \lambda_C^2 \cdot \varphi_B - \lambda_C^3$$



$$S(f, 0.1, 0.2) = \begin{bmatrix} 1 & 0.0854098872 & 2.4056003865 \cdot 10^{-3} \\ 2 & 0.0817853307 & 6.0871085192 \cdot 10^{-4} \\ 3 & 0.0805574829 & 7.7525675141 \cdot 10^{-6} \\ 4 & 0.0805416433 & 2.5537193679 \cdot 10^{-8} \\ 5 & 0.0805415909 & 1.0774159342 \cdot 10^{-12} \\ 6 & 0.0805415909 & 0 \end{bmatrix}$$

$$\varphi_B := 0.0805415909$$

$$\varphi_A := 3 \cdot \lambda_C^2 + 3 \cdot (1 - 2 \cdot \lambda_C) \cdot \varphi_B + \varphi_B^2 + 1 - 3 \cdot \lambda_C$$

$$\varphi_A = 0.4477392992326$$

$$b := \varphi_B \cdot \left( \frac{R \cdot T_C}{P_C} \right) = 0.0776357058388$$

$$F := 0.452413 + 1.30982 \, w - 0.295937 \, w^2$$

$$Tr := \frac{T}{(T_C)} = 1.0896708286039$$

$$F = 0.684601256175$$

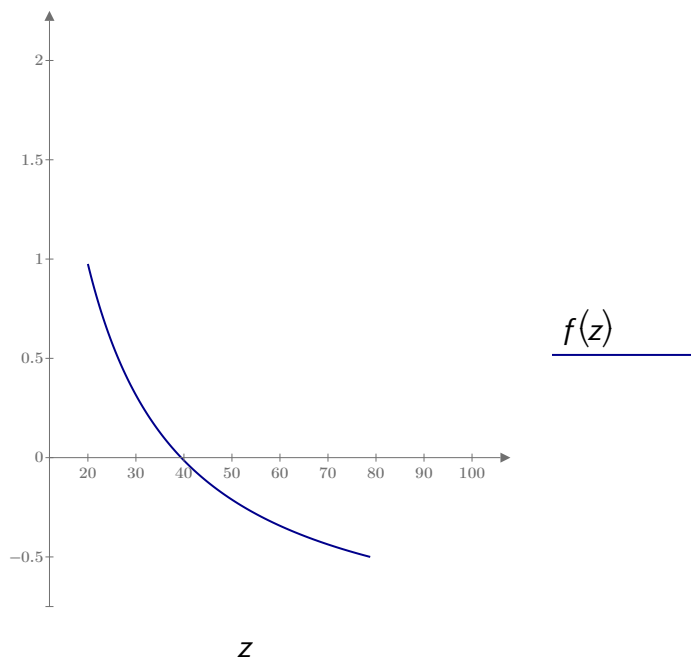


$$a := \varphi_A \cdot \left( \frac{R^2 \cdot T_C}{P_C} \right) \cdot \left( 1 + F \cdot \sqrt{T_r} \right)^2$$

$$a = 0.1040458187157$$

$$\frac{R \cdot T}{V - b} - \frac{a}{V \cdot (V + b) + c \cdot (V - b)} - P = 0$$

$$f(x) := \frac{R \cdot T}{x - b} - \frac{a}{x \cdot (x + b) + c \cdot (x - b)} - P$$



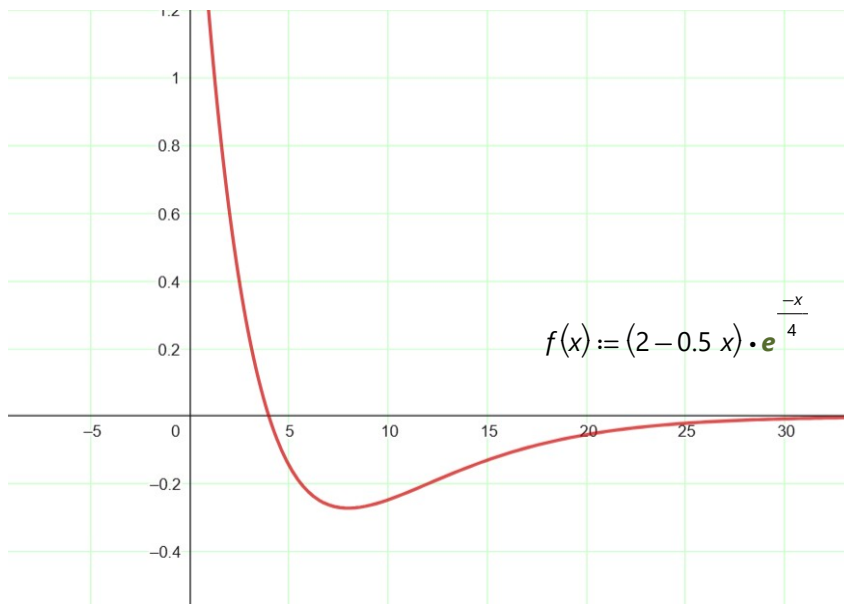
$$S(f, 20, 30) = \begin{bmatrix} 1 & 34.7762247734 & 0.1342543575 \\ 2 & 38.3182491035 & 0.0292015685 \\ 3 & 39.3028271158 & 3.3696503605 \cdot 10^{-3} \\ 4 & 39.4312606155 & 9.529524021 \cdot 10^{-5} \\ 5 & 39.4349984817 & 3.2000299921 \cdot 10^{-7} \\ 6 & 39.4350110758 & 3.0492053327 \cdot 10^{-11} \\ 7 & 39.435011077 & 0 \end{bmatrix}$$





$$\frac{d}{dt} (2t \cdot e^{-0.25t}) \rightarrow (-0.5 \cdot t + 2.0) \cdot e^{-0.25 \cdot t}$$

Utilizando el método de la Secante, determinar el tiempo que se requiere para que la concentración sea máxima



1° Iteración

$a := 1$

$b := 3$

$x_0 := 1$

$$c := b - \frac{(b-a) \cdot f(b)}{f(b) - f(a)}$$

$c = 3.5068213321168$

$k := 0 \dots 5$

$$x_{k+2} := x_{k+1} - \frac{(x_{k+1} - x_k) \cdot f(x_{k+1})}{f(x_{k+1}) - f(x_k)}$$

2° Iteración

$a := 3$

$b := c = 3.5068213321168$

$$c := b - \frac{(b-a) \cdot f(b)}{f(b) - f(a)}$$

$c = 3.8962152849512$

3° Iteración

$a := 3.5068213321168$

$b := c$



$$c := b - \frac{(b-a) \cdot f(b)}{f(b) - f(a)}$$

$$c = 3.9881011878492$$

$$S(f, 1, 3) = \begin{bmatrix} 1 & 3.5068213321168 & 0.1026185796243 \\ 2 & 3.8962152849512 & 0.0195919291667 \\ 3 & 3.9881011878492 & 0.0021951845015 \\ 4 & 3.9996956852009 & 0.0000559798378 \\ 5 & 3.999990961332 & 0.000000166257 \\ 6 & 3.9999999999312 & 0.0000000000126 \\ 7 & 4 & 0 \end{bmatrix}$$

$$k = \begin{bmatrix} 0 \\ 1 \\ 2 \\ 3 \\ 4 \\ 5 \end{bmatrix} \quad x_k = \begin{bmatrix} 1 \\ 3 \\ 3.50682133 \\ 3.89621528 \\ 3.98810118 \\ 3.99969568 \end{bmatrix}$$

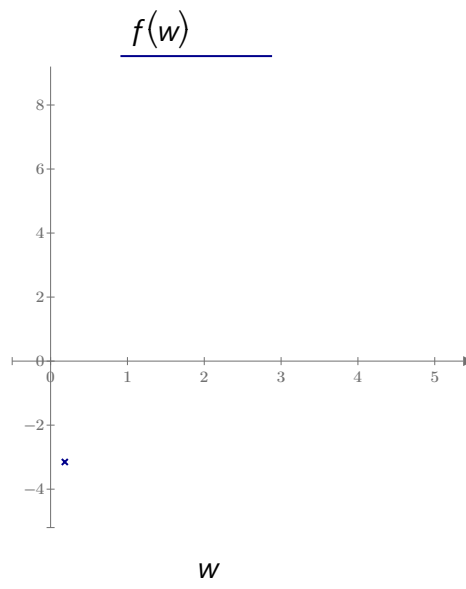
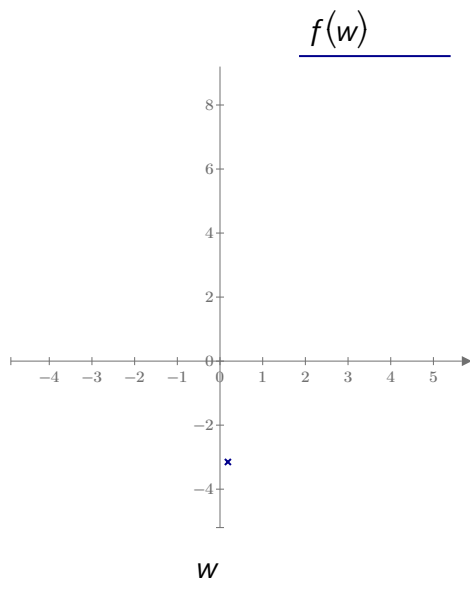
## Ejemplos

### Ejemplo 1

Hallar la solución positiva de

$$x^2 - x - 3 = 0$$

$$f(x) := x^2 - x - 3$$



$$x_0 := 2.5$$

$$x_1 := 3$$

$$k := 1..7$$

$$\begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

$$\begin{bmatrix} 3 \\ 2.3333333333333 \end{bmatrix}$$

$$x_{k+1} := x_k - \frac{f(x_k) \cdot (x_k - x_{k-1})}{f(x_k) - f(x_{k-1})}$$



$$k = \begin{bmatrix} 3 \\ 4 \\ 5 \\ 6 \\ 7 \end{bmatrix} \quad x_k = \begin{bmatrix} 2.3076923076923 \\ 2.3028169014085 \\ 2.3027756939235 \\ 2.3027756377326 \\ 2.302775637732 \end{bmatrix}$$

Con  $x_0$  y  $x_1$  saca  $x_2$

Con  $x_1$  y  $x_2$  saca  $x_3$

$$S(f, 2.5, 3) = \begin{bmatrix} 1 & 2.3333333333333 & 0.11111111111111 \\ 2 & 2.3076923076923 & 0.0177514792899 \\ 3 & 2.3028169014085 & 0.000148780004 \\ 4 & 2.3027756939235 & 0.0000002026013 \\ 5 & 2.3027756377326 & 0.0000000000023 \\ 6 & 2.302775637732 & 0 \end{bmatrix}$$

$$N_R(f, 3) = \begin{bmatrix} \text{" k "} & \text{" Valores de x"} & \text{" f ( x ) " } \\ 0 & 3 & 3 \\ 1 & 2.4 & 0.36 \\ 2 & 2.3052631578947 & 0.0089750692521 \\ 3 & 2.3027773515421 & 0.0000061792332 \\ 4 & 2.3027756377328 & 0.00000000000029 \\ 5 & 2.302775637732 & 0 \end{bmatrix}$$

$$N_{RM}(f, 3) = \begin{bmatrix} \text{" k "} & \text{" Valores de x"} & \text{" f ( x ) " } \\ 0 & 3 & 3 \\ 1 & 2.3181818181818 & 0.0557851239669 \\ 2 & 2.3027759154387 & 0.0000010012859 \\ 3 & 2.302775637732 & 0 \end{bmatrix}$$

## Ejemplo N° 2

Dada la siguiente ecuación encuentre el valor del factor de fricción (  $f$  ), si se sabe que el Número de Reynolds (Re) es 8200 y  $n = 0.8$

$$\frac{1}{f} = \frac{4}{n^{0.75}} \cdot \log(Re \cdot f^{1-0.5 \cdot n}) - \frac{0.4}{n^{1.2}}$$

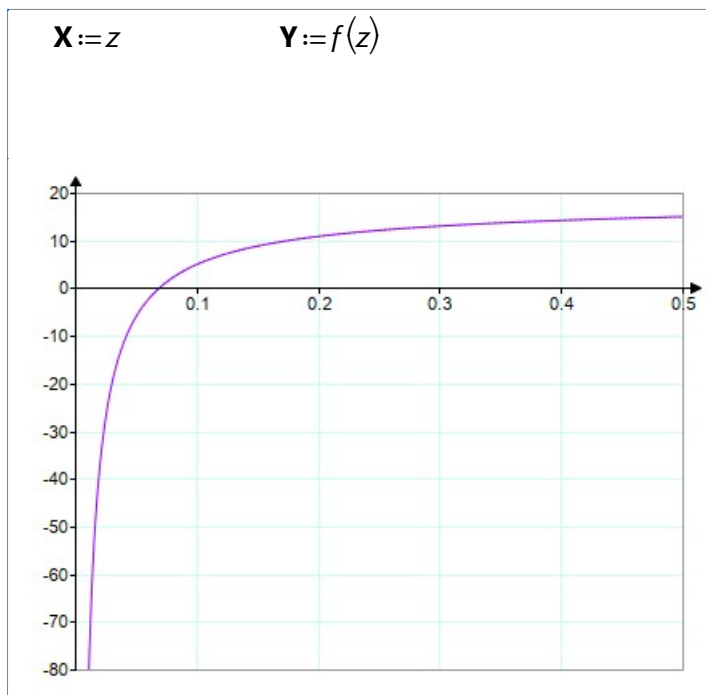
Datos $Re := 8200$  $n := 0.8$ haciendo  $x = f$ 

$$\frac{1}{x} = \frac{4}{n^{0.75}} \cdot \log(Re \cdot x^{1-0.5 \cdot n}) - \frac{0.4}{n^{1.2}}$$

$$0 = \frac{4}{n^{0.75}} \cdot \log(Re \cdot x^{1-0.5 \cdot n}) - \frac{0.4}{n^{1.2}} - \frac{1}{x}$$

$$f(x) := \frac{4}{n^{0.75}} \cdot \log(Re \cdot x^{1-0.5 \cdot n}) - \frac{0.4}{n^{1.2}} - \frac{1}{x}$$

$$d(x) := \frac{d}{dx} f(x)$$

 $z := 0.01, 0.011 \dots 0.5$ 

$$S(f, 0.05, 0.1) = \begin{bmatrix} 1 & 0.0762889447539 & 1.705696456585 \\ 2 & 0.0645372755817 & -0.887292767469 \\ 3 & 0.0685585692167 & 0.0960400696715 \\ 4 & 0.0681658178447 & 0.0049202887512 \\ 5 & 0.068144610044 & -0.0000287270654 \\ 6 & 0.0681447331468 & 0.0000000085441 \\ 7 & 0.0681447331102 & 0 \end{bmatrix}$$



$$N_R(f, 0.1) = \begin{bmatrix} \text{" k "} & \text{" Valores de x "} & \text{" f ( x ) " } \\ 0 & 0.1 & 5.1472384471563 \\ 1 & 0.0541742264923 & -4.0670136771089 \\ 2 & 0.0653633791381 & -0.6757857371815 \\ 3 & 0.0680353830877 & -0.025564713749 \\ 4 & 0.0681445644217 & -0.0000393765872 \\ 5 & 0.0681447331098 & -0.0000000000937 \\ 6 & 0.0681447331102 & 0 \end{bmatrix}$$

$$N_{RM}(f, 0.1) = \begin{bmatrix} \text{" k "} & \text{" Valores de x "} & \text{" f ( x ) " } \\ 0 & 0.1 & 5.1472384471563 \\ 1 & 0.0680238131612 & -0.0282742448618 \\ 2 & 0.0681447331195 & 0.0000000021648 \\ 3 & 0.0681447331102 & 0 \end{bmatrix}$$

3. Se desea determinar el diámetro de la tubería por donde circula un fluido a una velocidad de  $v = 11.11 \frac{ft}{s}$ , si se sabe que el factor de fricción obtenido mediante la correlación de Pavlov fue de  $f = 0.01986827$

$$f = \left\{ -2 \log \left[ \frac{1}{3.7} \left( \frac{\epsilon}{D} \right) + \left( \frac{6.81}{Re} \right)^{0.9} \right] \right\}^{-2}$$

Datos:

$$\rho := 59.4$$

$$\mu := 1.8815 \cdot 10^{-4}$$

$$V := 11.11$$

$$\epsilon := 0.5054 \cdot 10^{-3}$$

$$f := 0.01986827$$

$$Re = \frac{\rho \cdot V \cdot D}{\mu}$$

$$\left( -2 \cdot \log \left( \frac{1}{3.7} \cdot \left( \frac{\epsilon}{D} \right) + \left( \frac{6.81}{\frac{\rho \cdot V \cdot D}{\mu}} \right)^{0.9} \right) \right)^{-2} = f$$

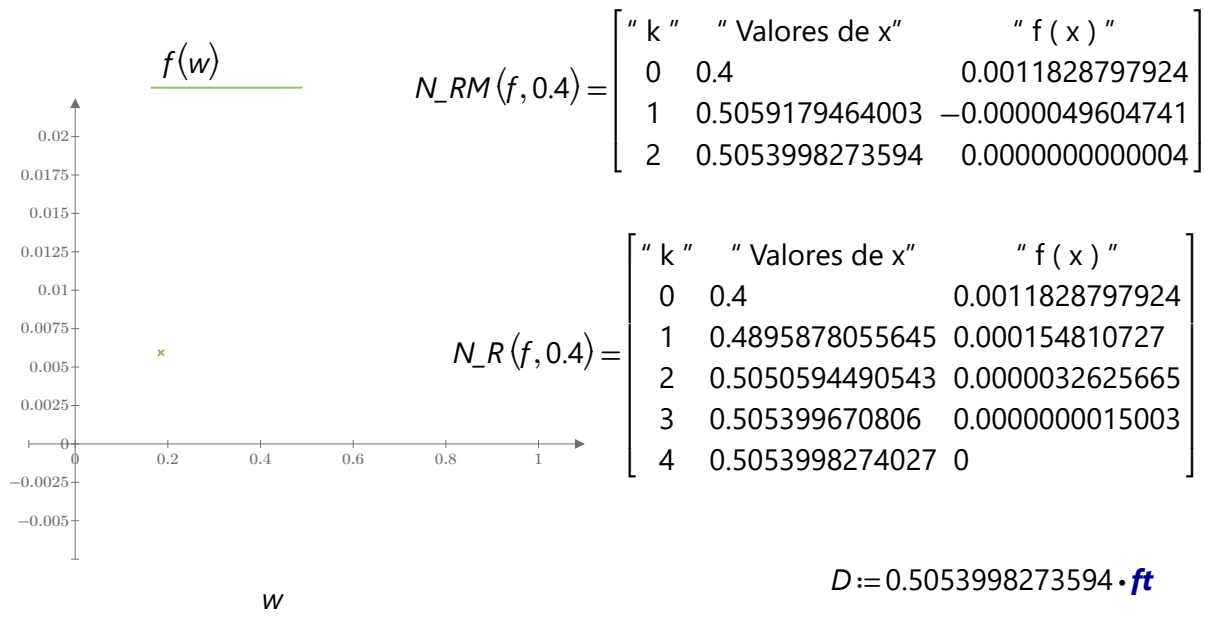
Número adimensional

$$\left( -2 \cdot \log \left( \frac{1}{3.7} \cdot \left( \frac{\epsilon}{D} \right) + \left( \frac{6.81 \cdot \mu}{\rho \cdot V \cdot D} \right)^{0.9} \right) \right)^{-2} - f = 0$$

$$\left( -2 \cdot \log \left( \frac{1}{3.7} \cdot \left( \frac{\epsilon}{x} \right) + \left( \frac{6.81 \cdot \mu}{\rho \cdot V \cdot x} \right)^{0.9} \right) \right)^{-2} - f = 0$$



$$f(x) := \left( -2 \cdot \log \left( \frac{1}{3.7} \cdot \left( \frac{\varepsilon}{x} \right) + \left( \frac{6.81 \cdot \mu}{\rho \cdot V \cdot x} \right)^{0.9} \right) \right)^{-2} - f$$



$$S(f, 0.2, 0.4) = \begin{bmatrix} 1 & 0.4563133597216 & 0.0005041229449 \\ 2 & 0.498138141082 & 0.0000702627918 \\ 3 & 0.5049115815159 & 0.0000046808303 \\ 4 & 0.505395027335 & 0.0000000459883 \\ 5 & 0.5053998242352 & 0.0000000000303 \\ 6 & 0.5053998274027 & 0 \end{bmatrix}$$

$$D := 0.5053998274027 \cdot ft$$

$$0.5053998274027 \cdot 30.48 = 15.4045867392343$$



Determinar las concentraciones  $Ca_1$  y  $Ca_2$  sabiendo que se lleva a cabo una reacción isotérmica e irreversible, verificándose las siguientes ecuaciones

$$F \cdot Cao + F_R \cdot Ca_2 - (F + F_R) \cdot Ca_1 = k \cdot V_1 \cdot Ca_1^n \quad \dots\dots\dots (1)$$

$$(F + F_R) \cdot Ca_1 - (F + F_R) \cdot Ca_2 = k \cdot V_2 \cdot Ca_2^n \quad \dots\dots\dots (2)$$

$$F := 25 \quad V_1 := 80 \quad V_2 := 20 \quad Cao := 1 \quad k := 0.2$$

$$F_R := 100 \quad n := 1.8$$

De (1)

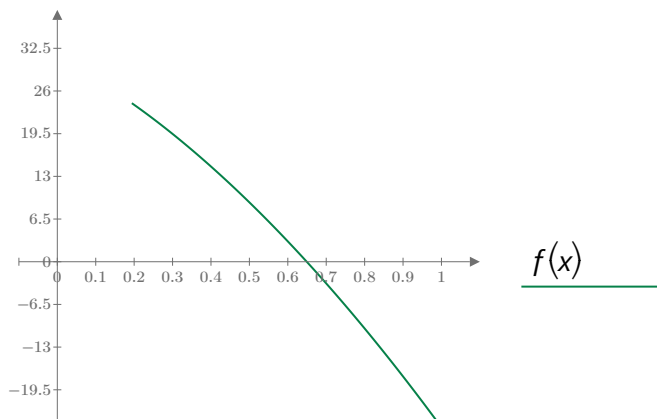
$$Ca_2 = \frac{1}{F_R} \cdot \left( k \cdot V_1 \cdot Ca_1^n + (F + F_R) \cdot Ca_1 - F \cdot Cao \right)$$

En (2)

$$(F + F_R) \cdot Ca_1 - (F + F_R) \cdot \left( \frac{1}{F_R} \cdot \left( k \cdot V_1 \cdot Ca_1^n + (F + F_R) \cdot Ca_1 - F \cdot Cao \right) \right) = k \cdot V_2 \cdot \left( \frac{1}{F_R} \cdot \left( k \cdot V_1 \cdot Ca_1^n + (F + F_R) \cdot Ca_1 - F \cdot Cao \right) \right)^n$$

$$f(x) := (F + F_R) \cdot x - \frac{(F + F_R)}{F_R} \cdot \left( k \cdot V_1 \cdot x^n + (F + F_R) \cdot x - F \cdot Cao \right) - \frac{k \cdot V_2}{F_R^n} \cdot \left( k \cdot V_1 \cdot x^n + (F + F_R) \cdot x - F \cdot Cao \right)^n$$

$$x := 0, 0.001 \dots 1$$





-26+

x

$$S(f, 0.2, 0.4) = \begin{bmatrix} 1 & 0.7086573768672 & -3.8805742991496 \\ 2 & 0.6434922447585 & 0.3708410864687 \\ 3 & 0.6491764478295 & 0.0077476298812 \\ 4 & 0.649297736442 & -0.0000161974537 \\ 5 & 0.649297483401 & 0.0000000007042 \\ 6 & 0.649297483412 & 0 \end{bmatrix}$$

$$N_R(f, 0.7) = \begin{bmatrix} \text{" k "} & \text{" Valores de x"} & \text{" f ( x ) " } \\ 0 & 0.7 & -3.3045867341082 \\ 1 & 0.6501828643171 & -0.056694760279 \\ 2 & 0.6492977647785 & -0.0000180113848 \\ 3 & 0.6492974834121 & -0.0000000000018 \\ 4 & 0.649297483412 & 0 \end{bmatrix}$$

$$N_{RM}(f, 0.7) = \begin{bmatrix} \text{" k "} & \text{" Valores de x"} & \text{" f ( x ) " } \\ 0 & 0.7 & -3.3045867341082 \\ 1 & 0.6493159273798 & -0.0011806792037 \\ 2 & 0.649297483412 & -0.0000000000001 \end{bmatrix}$$

$$Ca_1 := 0.649297483412 \frac{\text{mol}}{\text{L}}$$

$$Ca_2 := \frac{1}{F_R} \cdot \left( k \cdot V_1 \cdot Ca_1^n + (F + F_R) \cdot Ca_1 - F \cdot Cao \right)$$

$$Ca_2 = 0.6351610164859 \frac{\text{mol}}{\text{L}}$$